

API Integration Blueprint

Vision AI Prototype

1. Project Overview

This document explains where the API Integration module fits within the overall Vision AI system and clearly defines the responsibilities before, during, and after API integration.

2. Overall Workflow

User → Frontend → FastAPI Backend → Request Validation → API Integration Module → External AI Model → Image Processing → TIFF Conversion → Metadata Generation → Response → Frontend → User

3. Module Breakdown

Frontend

- UI
- Upload image
- Prompt input
- Calls backend endpoints
- Displays results

Backend Core

- Routes (/generate, /convert)
- Validation
- File handling
- Pipeline orchestration
- Response generation

My Responsibility - API Integration

Receive validated request from backend. Load API key securely. Create provider-specific request payload. Authenticate with AI provider. Send HTTPS request. Handle retries, timeouts, and HTTP errors. Receive generated image. Validate returned image. Normalize response into a common internal format. Return image back to backend pipeline.

Image Processing

- Upscaling
- Resize
- TIFF Conversion

- DPI Settings
- Metadata

4. Internal Module Flow

```
main.py
↓
generate.py
↓
image_generator.py (API Integration)
↓
External AI Provider (Pollinations / FLUX / OpenAI Images)
↓
Generated Image
↓
generate.py
↓
tiff_converter.py
↓
response_builder.py
↓
Frontend
```

5. Future Extensibility

Changing AI providers only requires modifying the API Integration module. Backend routes, frontend, and processing pipeline remain unchanged.

6. Deliverables

- AI API Integration
- Secure API Key Management
- Request/Response Mapping
- Error Handling
- Retry Logic
- Image Retrieval
- Return standardized image object to backend